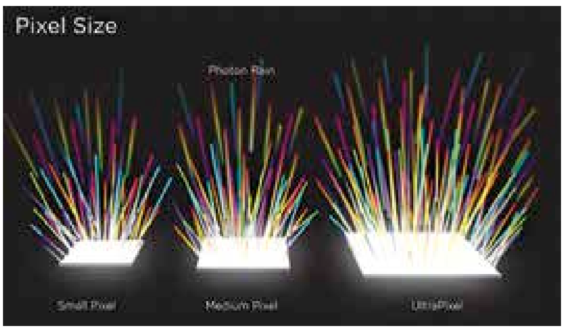
Ford's Mustang-esque Electric SUV
Ford confirmed that their all-electric SUV "inspired" by the Mustang design language will be revealed this year. **https://dgit.in/feb19-79**

Toshiba brings UFS 3.0 based chip
The industry's first embedded flash chips to use the UFS version 3.0 which are targeted towards portable devices. **https://dgit.in/feb19-80**

PIXEL ARRANGEMENT

Another area of innovation is how the pixels are arranged in the matrix, known as the mosaic. Now, the traditional arrangement was what is known as the bayer filter, which used 2x2 blocks, each with two green pixels, one blue pixel and one red pixel. There are twice the number of green pixels, as the human vision is more sensitive to that colour. The latest sensors from both Samsung and Sony, and very probably the Huawei sensor in the P20, use what is known as a quad bayer filter. Samsung tags its products that use the arrangement with the name "tetracell". In this type of sensor, there are 8x8 blocks, with 2x2 blocks arranged in the bayer arrangement. This allows the four neighbouring pixels to behave like one large pixel, improving the dynamic range in images, and allowing for better low light performance. The signals from the adjacent pixels are added in low light conditions, by essentially boosting the size of a single "pixel" from say 0.8 µm to 1.6 µm. The new quad bayer arrangement is even more effective when combined with exposure control technologies and signal processing. In low light situations, the quad bayer filter reduces the amount of noise, but the images are of a lower MP count.



CREDIT: HTC

How pixel size matters

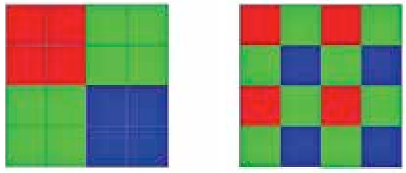
FOCUSING TECHNOLOGIES

Another job the sensor has to do is to focus the image, and relay the information to a motor that adjusts the lens accordingly. The oldest and cheapest method is contrast AF, which was a time consuming process. The lens would continuously move till the sensor found the sharpest image. This technology is known as CDAF, and is found in older and cheaper phones. CDAF fails at focusing moving objects, and does not work very well for videos. Phase detection does a lot of work in a very short amount of time, improving the speed of the AF. PDAF takes two slightly different images, compares the two for differences in sharpness, calculates the direction and distance by which the lens has to move to achieve the sharpest image, then moves the lens. PDAF is faster, and works well with moving subjects and during video shoots. An increasing number of smartphone cameras have been offering PDAF since 2017.

The problem with PDAF is that a certain amount of the pixels in the array (about 5 percent) are replaced with the PD diodes, also known as PD pixels. Samsung has an innovative solution on its sensors known as Dual PD, where every pixel in the array is accompanied by its own PD diode. This allows for much more faster focusing. PDAF also does not work as well in low light situations. Some cameras, such as the Pixel, use a laser to ascertain the distance to the subject, which allows for quick AF even in low light conditions.

LENSES

The first dual lens smartphone, the HTC Evo 3D was for capturing stereoscopic images. Dual lens cameras have become mainstream now, with an additional camera offering either telephoto or wide angle capabilities. This forced consumers to pick one of the two, which is why 2018 saw an influx of triple camera smartphones. The first to sport a triple camera setup was the Huawei P20, which had a telephoto camera but with a monochrome sensor to support the main camera. This was a trick that Huawei used in the P10 as well, where images from a high resolution black and white sensor is combined with a lower resolution colour image, to produce sharp and detailed images, two reasons why some photography professionals prefer monochrome sensors. The LG V40 offers the most flexibility to the users, with a telephoto lens and a wide angle lens apart from the main camera. The Samsung A9 went a step further, throwing in a depth sensor and offering a 4-camera setup.



Quad bayer arrangement on the left, the bayer filter on the right

Google seems to be the only manufacturer holding out, stubbornly using only one camera on its Pixel devices. Apple seems to be jumping onto the bandwagon too, with leaked renders of its upcoming flagship showing a three camera setup. There are also leaked images of a Nokia devices with five cameras on the back. LG has a patent for a 16 camera setup for capturing "Moving Images", perhaps similar to Apple's "Live Photos", but it's unclear if this will be realised in a product.

There have been some innovative approaches for solving the problem of a lack of optical zoom in smartphone cameras. The Oppo 5X had a periscope like setup, where the sensor moved sideways within the body of the camera, instead of a protruding telescopic lens.

Once the images are captured by the hardware, the software gets to work. Smartphones can get the much desired bokeh effect to work with a single lens, cameras can identify images while shooting to tweak parameters for a better shot, image signal processing is getting increasingly smarter, and artificial intelligence has a major role to play in enhancing images in various ways. Make no mistake, there is a lot of innovation in the software department too, when it comes to smartphone photography, but that is another story.